

# MyHourly - Scheduler & Transactional Performance Fixes

Issue report and fixes applied - August 2026

## 1. Summary

All scheduler and @Transactional code paths were reviewed for performance issues. Six problems were found: an N+1 notification fan-out (with two DB round trips per recipient), in-memory filtering of every employee, redundant work in the 1-minute and 5-minute jobs, and per-combination queries in the monthly/yearly leave jobs. All fixes are behavior-preserving: notification dedup semantics, messages, titles, priorities, cron schedules and transaction boundaries are unchanged. The project compiles cleanly. It was NOT started against a live database, because the dev profile points at a production Aiven PostgreSQL instance with `ddl-auto=update`; verify on local/staging first (see `SCHEDULER_PERFORMANCE_VERIFICATION.md`).

## 2. Issues Found

| # | Issue                                                                                                                                                                          | Impact                                                                                                                     |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1 | Spring Boot default task scheduler is single-threaded; all @Scheduled jobs share one thread.                                                                                   | Long daily notification runs block the 1-minute and 5-minute jobs, delaying missed-checkout marking and notifications.     |
| 2 | Birthday / Work-anniversary schedulers load every active employee and filter in memory, then call createNotification (exists-check + save) in nested loops over all employees. | O(birthdays x employees) x 2 DB round trips; a long-running transaction holding a pooled connection while other jobs wait. |
| 3 | 5-minute processor scans all of today's attendance and recent leave rows every run and issues 2 queries per row, all day long.                                                 | Recurring N+1 load every 5 minutes; most queries re-check already-notified rows.                                           |
| 4 | Announcement broadcast calls createNotification per employee.                                                                                                                  | 2 x employees queries per announcement posted.                                                                             |
| 5 | markMissedCheckouts (every 60 s) re-fetches and re-saves rows already marked MISSED_CHECKOUT after the daily cutoff.                                                           | Redundant select + saveAll every minute for the rest of the day.                                                           |
| 6 | Leave expiry and leave allocation issue per (employee x leave type) balance / used-days queries; allocation also re-fetches leave types per employee.                          | Thousands of queries in the month-end and yearly jobs; long transactions under load.                                       |

## 3. Fixes Applied

### 3.1 Multi-threaded task scheduler

New SchedulingConfig registers a ThreadPoolTaskScheduler (pool size 4, graceful shutdown, error handler that logs 'Scheduled task failed'). Spring Boot's default scheduler is single-threaded, so a long daily notification run used to block the 1-minute missed-checkout check and the 5-minute notification processor. Jobs now run in parallel.

### 3.2 Bulk notification creation

NotificationRepository gained a batch duplicate-check query returning existing (employeeId, referenceId) pairs. NotificationService gained createNotificationsBulk(...) and a NotificationItem record: one exists-check query + one saveAll per (referenceType, notificationType) group, with the same first-wins dedup as the old per-call createNotification (employeeId + referenceType + referenceId + notificationType).

### 3.3 Birthday / Work-anniversary schedulers

Birthdays and anniversaries are now filtered in SQL via HQL MONTH()/DAY() instead of loading every active employee and filtering in Java (verified the functions are registered in the exact Hibernate 7.2.12 on the classpath). 'Notify everyone else' now uses bulk creation: O(2) queries per birthday employee instead of 2 x employees.

### 3.4 Holiday scheduler

Holiday-today and tomorrow-reminder fan-outs use bulk creation: 2 queries total instead of 2 x employees. Message text, priority and dedup are identical.

### 3.5 5-minute notification processor

processAttendanceNotifications now queries only LATE / ABSENT / MISSED\_CHECKOUT rows and builds all notifications in one bulk call (employee + reporting-manager items). processLeaveNotifications and processCheckoutReminderNotifications are batched the same way. Exact titles/messages/priorities preserved, including the existing 'Attendance' string used as the manager-notification message.

### 3.6 Announcement broadcast

saveAnnouncementAndBroadcast builds one bulk batch for all employees instead of calling createNotification per employee.

### 3.7 Missed-checkout marker (every 60 s)

markMissedCheckouts uses a narrowed query that excludes rows already marked MISSED\_CHECKOUT or LEAVE, so after the first run the minute-by-minute job does zero re-fetching / re-saving. Same outcome as before.

### 3.8 Leave expiry (month-end)

expireMonthlyUnused now loads all year balances (findByYear) and a grouped used-days query (new sumApprovedLeaveDaysInMonthGrouped projection) in two reads, replacing 2 x employees x leaveTypes per-combination queries. The expiry math (used days, unused guideline, clamping) is unchanged.

### 3.9 Leave allocation (yearly / all-employee)

The batched allocation path loads active leave types and the year's existing balances once, instead of re-fetching leave types per employee and balance per combination. Creation and adjustment logic is unchanged. New-employee onboarding and per-leave-type reallocation endpoints keep their original per-record lookups.

## 4. Files Changed

| File                                                   | Change                                                                                                      |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| common/config/SchedulingConfig.java                    | NEW - 4-thread task scheduler with graceful shutdown and error logging.                                     |
| notification/repository/NotificationRepository.java    | Added batch duplicate-check query (employeeId, referenceId pairs).                                          |
| notification/service/NotificationService.java          | Added createNotificationsBulk(...) and NotificationItem record.                                             |
| notification/service/impl/NotificationServiceImpl.java | Bulk implementation; batched attendance/leave/checkout-reminder processing; batched announcement broadcast. |
| notification/scheduler/BirthdayScheduler.java          | SQL birthday filter + bulk notification creation.                                                           |
| notification/scheduler/WorkAnniversaryScheduler.java   | SQL anniversary filter + bulk notification creation.                                                        |
| notification/scheduler/HolidayScheduler.java           | Bulk notification creation for holiday + reminder.                                                          |
| employee/repository/EmployeeRepository.java            | Added findActiveEmployeesWithBirthday / findActiveEmployeesWithWorkAnniversary (HQL MONTH()/DAY()).         |

|                                                    |                                                                      |
|----------------------------------------------------|----------------------------------------------------------------------|
| attendance/repository/AttendanceRepository.java    | Added status-filtered query for the missed-checkout marker.          |
| attendance/service/impl/AttendanceServiceImpl.java | markMissedCheckouts now skips already-marked rows via the query.     |
| leave/repository/LeaveRequestRepository.java       | Added grouped used-days projection query for month-end expiry.       |
| leave/service/impl/LeaveExpiryServiceImpl.java     | Two batched reads replace the per-combination N+1; same expiry math. |
| leave/service/impl/LeaveAllocationServiceImpl.java | Batched allocation path loads leave types + year balances once.      |

## 5. Verification & Limitations

- o mvnw compile passes (no compile errors; only the usual Lombok warning).
- o HQL day()/month() functions confirmed present in Hibernate 7.2.12 CommonFunctionFactory (the version resolved by Spring Boot 4.0.7).
- o NOT verified at runtime: app startup and tests were not run against a database, to avoid touching the live Aiven PostgreSQL configured in the dev profile.
- o Parallel scheduler caveat: a job that takes longer than its own interval could now overlap itself; after these optimizations each run is milliseconds, so this is not a practical risk, and the notification dedup check is not race-proof.
- o No schema, API, cron schedule, or transaction-boundary changes were made.

## 6. Suggested Follow-ups

- o Batch the remaining per-row saves in leave allocation / leave expiry (saveAll for balances and transactions).
- o Add a unique DB index on notifications(employee\_id, reference\_type, reference\_id, notification\_type) to speed the duplicate checks and make them race-proof.
- o Add EXPLAIN / query-plan checks for the new MONTH()/DAY() and grouped leave queries on PostgreSQL.